

Harshitha Palaram

📍 Bengaluru, Karnataka, India ✉️ harshithapalaram09@gmail.com 📱 8310419482 🌐 in/LinkedIn

SUMMARY

Highly motivated and skilled Artificial Intelligence and Data Science student with a strong foundation in machine learning, deep learning, and computer vision. Seeking to leverage practical experience in developing and optimizing predictive models, implementing advanced AI algorithms, and contributing to innovative projects. Proven ability to architect complex deep learning pipelines, achieve top rankings in competitive challenges, and develop end-to-end AI-powered applications.

EXPERIENCE

Machine Learning Intern

Cognifyz IT Solutions Pvt. Ltd.

April 2026 – May 2026, Remote

- Engineered and evaluated end to end machine learning pipelines for restaurant rating prediction, recommendation systems, cuisine classification, and location analytics using Python, Pandas, NumPy, Scikit-learn, TF-IDF, and cosine similarity, strengthening predictive modeling and data analysis workflows.
- [Certificate]
- Optimized regression models using Random Forest, XGBoost, and CatBoost, achieving 95.35% R^2 with 0.33 RMSE; leveraged DBSCAN clustering and SHAP explainability to extract actionable insights and improve model interpretability.

PROJECT

Glacial Lake Outburst Flood Detection (GLOFeagles'26)

GitHub

- Architected a dual-task deep learning pipeline using Efficient Net-B4 for 6-class classification and U-Net++ for semantic segmentation of glacial lakes from satellite imagery.
- Achieved 0.6260 Mean IoU, 0.6920 Dice/F1, and 0.7656 Precision on 2,220 image-mask pairs using hybrid Focal + Tversky Loss, mixed precision training (AMP), and Albumentations.

NTIRE 2026 Rip Current Detection & Segmentation Challenge

GitHub

- Designed a YOLOv8-based computer vision pipeline for rip current detection using segmentation, contour extraction, morphological operations, and post-processing for accurate localization.
- Implemented multi-scale inference and test-time augmentation (TTA), contributing to a Top 7 global ranking among 900+ submissions at the NTIRE CVPR 2026 Workshop Challenge.

FinAscend: AI-Powered Financial Decision System

GitHub

- Developed an AI powered financial analytics platform integrating OCR and API pipelines to process invoices and financial statements for small businesses.
- Implemented Monte Carlo simulations, probabilistic risk forecasting, and Explainable AI techniques to predict liquidity, runway, and bankruptcy risk.

Learn-Lynx: GenAI-Powered Study Partner

GitHub

- Created a Retrieval-Augmented Generation (RAG) study assistant using LangChain, Sentence Transformers, and Hugging Face for semantic search, summarization, and question answering.
- Delivered an interactive Streamlit application providing personalized learning plans and real-time AI-assisted study support through vector-based document retrieval.

Food Delivery Web Application

GitHub | Live Demo | Admin

- Built a full-stack food ordering platform using React.js, Node.js, Express.js, and MongoDB featuring authentication, cart management, and order processing.
- Implemented JWT authentication, role-based access control, RESTful APIs, and deployed production-ready applications on Vercel and Render.

EDUCATION

B.Tech in Artificial Intelligence and Data Science

Shiv Nadar University • Chennai • May 2027 • 7.713
• CGPA: 7.713.

Intermediate (Class XII)

FIITJEE Junior College • Hyderabad • May 2021
• 95.4%.

Class X (CBSE)

Royale Concorde International School • Bengaluru • May 2021
• 96.6%.

CERTIFICATIONS

Affective Computing

NPTEL • January 2026

Programming in Java

NPTEL • July 2025

Machine Learning Specialization

DeepLearning.AI • February 2025

AWARDS & HONORS

Top 5 Finalist - GLOFeagles Challenge

NCVPRIPG 2026 • 2026
• Selected among the Top 5 teams and invited to present the solution at the NCVPRIPG 2026 finale in LNMIIT, Jaipur.

Top 7 Global Rank NTIRE 2026 Rip Current Detection & Segmentation Challenge

CVPR • 2026
• Secured 7th place among 900+ global submissions in an international computer vision challenge.

Leaderboard Finalist SNUC Hacks 26 FinTech Hackathon

2026
• Presented FinAscend, an AI-powered financial decision support system, and advanced to the hackathon finals.

SKILLS

Python, Java, C, JavaScript (ES6+), SQL, HTML/CSS
React.js, Node.js, Express.js, FastAPI, REST APIs, JWT, Tailwind CSS
PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLOv8, Hugging Face Transformers, Sentence Transformers, RAG, Semantic Segmentation
NumPy, Pandas, Matplotlib, NLTK
MongoDB, MySQL
